Om Kumar

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Summary

Final-year B.Tech (Mathematics and Computing) student with hands-on software engineering and applied AI/ML experience: built and deployed ML models and CI/CD-driven data pipelines, shipped real-time perception and navigation software on a robotics team (3rd place, IGVC 2025), and contributed to open-source Python/Kubernetes projects and ran tests on AWS. Comfortable with Git/GitHub workflows, Docker, and cross-functional collaboration.

Education

Manipal Institute of Technology — Manipal, Karnataka, India
B.Tech in Mathematics and Computing — *July 2023 – 2027 (Expected)*

- CGPA: 8.33 / 10.0
- Relevant Coursework: Data Structures and Algorithms, Agile/Scrum Methodologies, Machine Learning, Deep Learning, Statistics, Probability Theory, Computational Linear Algebra, Optimization, Real Analysis, Industrial Robotics.

Technical Skills

Programming Languages: Python, C++, Java, SQL, JavaScript
AI / Machine Learning: PyTorch, TensorFlow, Deep Learning, Reinforcement Learning, Fine-Tuning, Computer Vision, Diffusion Models, 3D Gaussian Splatting, Neural Rendering, ONNX, OpenCV
Software Engineering Tools: Git, GitHub, GitHub Actions (CI/CD), AWS, Docker, Kubernetes, Linux, REST-style web apps (Node.js, Next.js, React, Supabase), Selenium, ROS/ROS2, CUDA
Practices: Agile/Scrum, CI/CD pipelines, open-source collaboration, pull requests and code review, experiment tracking and data versioning, reproducible ML pipelines

Experience

Machine Learning Research Intern — July 2026 – Present
Max Planck Institute for Intelligent Systems — *Tübingen, Germany*

- Part of an international research cohort of 12 people from 9 countries.
- Built scalable data generation pipelines for image-to-3D reconstruction, following CI/CD practices (GitHub Actions and Gitlab) for automated, repeatable runs.
- Trained end-to-end deep learning pipelines on gigabyte-scale 3D datasets, outperforming SOTA by **40%**.
- Improving pipeline robustness for real-world deployment.

Netweb Summer Research Intern — May 2025 – May 2026
NISER Bhubaneswar (Hybrid) — *India*

- Developed a post hoc floater removal pipeline for 3D Gaussian Splatting that removes spurious primitives without retraining; reduced model size to **20.4%** of the original Gaussian count while improving reconstruction quality by **+4.75 dB PSNR**.
- Developed a multi-view diffusion framework for novel view synthesis in sparse regions, achieving up to **+10% PSNR**, **+15% SSIM**, and **−20% LPIPS** over baselines.

Research Intern — Nov 2024 – May 2026
KLIV Group, IIT Kharagpur (Remote) — *India*

- Co-developed an arbitrary-scale super-resolution model for ultrasonography, outperforming bicubic and baseline CNNs on PSNR/SSIM.
- Co-developed an ultra-lightweight segmentation model that beats SOTA by about **7.5%**.
- Built a reproducible, production-ready training pipeline with automated data versioning and experiment tracking, supporting experiments across **100K+** medical images.

AI Subsystem Team Member — Mar 2024 – Present
Project MANAS (Autonomous Vehicles) — *Manipal, India*

- Placed **3rd globally** at IGVC 2025 (USA), AutoNav Driverless Challenge, with a fully autonomous ground vehicle.
- Developed and deployed a real-time object detection, localization and classification pipeline with **sub-30 ms** inference on embedded hardware using ONNX-optimized models.
- Built an end-to-end autonomous navigation stack (perception, path planning, control) on ROS2.
- Designed a probabilistic occupancy grid mapping framework that cut memory overhead by **40%** for real-time decision-making.

Research Assistant Mar 2024 – Sept 2024
Manipal Institute of Technology *Manipal, India*

- Applied Generative Flow Networks to generate novel drug candidates for *S. aureus*, expanding the candidate space beyond rule-based baselines.

Part-Time Teacher (Mathematics and Computer Science) 2019 – Present
Gurukul Classes (Remote)

- Deliver 1-on-1 mathematics and computer science instruction to international students with **100%** client retention.

Projects

Project Tirtha (Open Source) | Python, PyTorch, Nerfstudio GitHub PR

Integrated VGGT-based reconstruction for NISER's heritage research platform, giving about **1000**× faster inference than the previous pipeline; migrated the viewer to the PlayCanvas engine to improve deployment.

SREGym (Open Source) | Python, Kubernetes GitHub

Contributing fault scenarios to SREGym, an agentic evaluation framework for site reliability engineering (UIUC++ summer program); ran Kubernetes clusters on AWS nodes for testing.

Gurukul Tutoring Platform | Node.js, Next.js, React, Supabase, Vercel GitHub

Built and deployed a full-stack tutoring platform with online onboarding, extending the service beyond local geography.

Automated Tender Acquisition (GeM) | Python, Selenium GitHub

Automated government tender submissions, reducing manual effort from hours to minutes per tender.

Monocular Depth Estimation for Terrain Mapping | PyTorch, OpenCV GitHub

Generated depth from UAV RGB imagery and integrated it with path planning for autonomous exploration (ISDC 2025).

Ultrasound Image Super-Resolution | PyTorch GitHub

Developed a lightweight attention-based U-Net that improves diagnostic clarity over standard interpolation.

Leadership and Activities

Team Manager Aug 2025 – Present
Project MANAS, MIT Manipal *Manipal, India*

- Lead research, outreach, budgeting and operations for the university's AI and robotics research team.
- Designed recruitment challenges for new members, structured around CI/CD pipelines on GitHub Actions (github.com/Project-MANAS-Research-AI).

Associate Oct 2023 – Sept 2024
Entrepreneurship Cell, MIT Manipal *Manipal, India*

- Helped organize Manipal Entrepreneurship Summit 2024; designed reports and newsletters.

Local Committee Member Oct 2023 – Sept 2024
IAESTE *Manipal, India*

- Hosted international interns and produced content for conferences.

Languages

English (B2), Spanish (A1)